

IHARQ Layer 0 Claim Governance Through P02

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Authority boundary. Layer 0 governs claim wording, scope, qualification and prohibition. It does not change measurements, counts, metrics, run inclusion, source evidence, Protocol records or the finalized Phase Analysis.

Executive governance decision

The current Layer 0 state contains **37 reviewed claims: 2 APPROVED, 33 APPROVED_WITH_QUALIFICATIONS, 1 DEFERRED, and 1 REJECTED**. The 19 P00/P01 reviewed claims are preserved historically; 10 P02 candidate claims and 8 cumulative P00-P02 claims are newly reviewed under the same taxonomy.

P02 does not erase earlier decisions. In particular, P01-CLM-011/v1 remains historically **DEFERRED** because P01 did not test A4 effectiveness. P02 later supplies governed evidence that does not support a generic C1-C3 improvement under the executed constrained scope; the current resolution is carried by P02-CLM-007/v1 and CUM-CLM-005/v1, not by retroactive rewriting of the P01 disposition.

Governance invariants

- P00/P01 historical claim IDs, reviewed wording and dispositions remain preserved.
- All 10 P02 candidate claims are dispositioned.
- All 8 cumulative candidate claims are dispositioned.
- Stage18S remains post-hoc/descriptive/sensitivity evidence.
- A4 resource restrictions remain mandatory claim ceilings.
- Diagnostic Stage11 evidence does not replace primary A0.
- Technical readiness is not deployment or clinical readiness.
- Negative/null evidence is governed with the same scope discipline as positive evidence.

P02 reviewed claims

P02-CLM-001/v1 - APPROVED_WITH_QUALIFICATIONS

Source candidate: P02-CLAIM-CAND-001

Reviewed wording: Under the frozen P01 subject-grouped binary motor-imagery protocol, P02 established a complete multi-family raw accept-all decoder/baseline measurement spine across the three activated public EEG datasets; this is Layer-2 measurement evidence and does not establish calibration, abstention, clinical effectiveness, safety, or deployment readiness.

Evidence class: LAYER2_MEASUREMENT_CLOSURE

Claim ceiling: SUPPORTED

Supporting findings: P02-FIND-001

Numeric support: {"AO_terminal_cells": 678, "SUCCESS": 657, "FAILED": 3, "CONDITIONAL_SKIP": 15, "DEPENDENCY_BLOCKED": 3, "datasets": 3}

Mandatory qualification: Raw accept-all decoder evidence only; not clinical or safety evidence.

Prohibited stronger wording: Do not call P02 calibration-, threshold-, abstention-, safety-, or deployment-ready.

Post-hoc/diagnostic status: NO

P02-CLM-002/v1 - APPROVED_WITH_QUALIFICATIONS

Source candidate: P02-CLAIM-CAND-002

Reviewed wording: Within full-training A0, descriptive decoder ordering was dataset-dependent: CSP-LDA led the single-participant BNCI2014_001 test split, while DBConformer had the highest participant-first mean BACC on Lee2019_MI and PhysioNetMI. These rankings are descriptive and do not establish a universal winning decoder.

Evidence class: DESCRIPTIVE_MODEL_HETEROGENEITY

Claim ceiling: SUPPORTED_WITH_QUALIFICATION

Supporting findings: P02-FIND-002, P02-FIND-003

Numeric support: {"full_training_descriptive_leaders": {"BNCI2014_001": {"branch_id": "CLS-CSP-LDA", "participant_first_mean_BACC": 0.6388888888888888, "participants": 1}, "Lee2019_MI": {"branch_id": "DNN-SEQ", "participant_first_mean_BACC": 0.795, "participants": 6}, "PhysioNetMI": {"branch_id": "DNN-SEQ", "participant_first_mean_BACC": 0.649724517906336, "participants": 11}}}

Mandatory qualification: Descriptive rankings; no universal winner; BNCI inferential $n=1$.

Prohibited stronger wording: Do not claim one decoder is universally best across datasets or participants.

Post-hoc/diagnostic status: NO

P02-CLM-003/v1 - APPROVED_WITH_QUALIFICATIONS

Source candidate: P02-CLAIM-CAND-003

Reviewed wording: On PhysioNetMI full-training A0, the governed Friedman comparison detected overall decoder heterogeneity (statistic 20.544, $p=0.00451$, Kendall $W=0.267$), while no individual alternative-versus-CSP-LDA contrast survived Holm correction; the omnibus result therefore does not identify a statistically supported pairwise winner.

Evidence class: GOVERNED_STATISTICAL_RESULT

Claim ceiling: SUPPORTED_WITH_QUALIFICATION

Supporting findings: P02-FIND-003

Numeric support: {"PhysioNetMI_Friedman_statistic": 20.544, "p_value": 0.00451, "Kendall_W": 0.267, "Lee2019_MI_Friedman_p": 0.148, "Holm_significant_alt_vs_CSP_LDA": 0}

Mandatory qualification: Global omnibus heterogeneity is not evidence that a specific decoder is superior.

Prohibited stronger wording: Do not convert omnibus heterogeneity into a significant pairwise winner claim.

Post-hoc/diagnostic status: NO

P02-CLM-004/v1 - APPROVED_WITH_QUALIFICATIONS

Source candidate: P02-CLAIM-CAND-004

Reviewed wording: Across the governed 1, 2, 4, 8, 16 and 32 labels/class P02 low-label surface, performance was difficult, non-monotonic and rank-changing under one frozen subset per budget; the evidence does not support a stable or universal sample-efficiency ordering among the evaluated branches.

Evidence class: DESCRIPTIVE_LOW_LABEL

Claim ceiling: SUPPORTED_WITH_QUALIFICATION

Supporting findings: P02-FIND-005

Numeric support: {"governed_low_label_budgets_per_class": [1, 2, 4, 8, 16, 32], "leader_changes": {"BNCI2014_001": 4,

"Lee2019_MI": 4, "PhysioNetMI": 6}, "design": "single frozen subset per budget"}

Mandatory qualification: One frozen subset per budget; descriptive only.

Prohibited stronger wording: Do not claim stable sample-efficiency ordering or subset-robust low-label behavior.

Post-hoc/diagnostic status: NO

P02-CLM-005/v1 - APPROVED_WITH_QUALIFICATIONS

Source candidate: P02-CLAIM-CAND-005

Reviewed wording: The preserved P02 training-correction history shows that model viability and optimization behavior under the implemented configurations depended materially on training-policy choice. This supports treating inadequate optimization as distinct from demonstrated architectural incapacity, but does not establish universal test-set gains, architecture superiority, or theoretical invalidity of the earlier generic policy.

Evidence class: METHODOLOGICAL_CORRECTION

Claim ceiling: SUPPORTED_WITH_QUALIFICATION

Supporting findings: P02-FIND-006, P02-FIND-008

Numeric support: {"Stage12_validation_only_examples": [{"dataset_id": "BNCI2014_001", "branch_id": "DNN-FBCNET", "A_BACC": 0.5, "B_BACC": 0.7222222222222222, "absolute_delta_BACC": 0.2222222222222222, "claim_bearing": false}, {"dataset_id": "PhysioNetMI", "branch_id": "DNN-FBCNET", "A_BACC": 0.6395092565383486, "B_BACC": 0.6601038299539622, "absolute_delta_BACC": 0.0205945734156136, "claim_bearing": false}, {"dataset_id": "PhysioNetMI", "branch_id": "DNN-SEQ", "A_BACC": 0.6985502987559996, "B_BACC": 0.7167940052894505, "absolute_delta_BACC": 0.0182437065334508, "claim_bearing": false}, {"dataset_id": "PhysioNetMI", "branch_id": "SSL-CBRAMOD", "A_BACC": 0.5, "B_BACC": 0.6384072876873348, "absolute_delta_BACC": 0.1384072876873348, "claim_bearing": false}], "test_role_loaded": false}

Mandatory qualification: Mechanistic interpretation; source diagnostics are not a direct claim of test-set performance improvement for every model.

Prohibited stronger wording: Do not claim architecture superiority or theoretical invalidity of the original uniform policy.

Post-hoc/diagnostic status: NO

P02-CLM-006/v1 - APPROVED_WITH_QUALIFICATIONS

Source candidate: P02-CLAIM-CAND-006

Reviewed wording: The Stage-11 S&R diagnostic challenger did not yield a consistent participant-level BACC improvement over the canonical primary EEGNet across BNCI2014_001, Lee2019_MI and PhysioNetMI; it remains diagnostic-only and does not replace primary A0 evidence.

Evidence class: DIAGNOSTIC_NEGATIVE_RESULT

Claim ceiling: SUPPORTED

Supporting findings: P02-FIND-007

Numeric support: {"diagnostic_challenger": [{"dataset_id": "BNCI2014_001", "n": 5, "median_delta_BACC": -0.0347222222222222, "ci_low": -0.0763888888888889, "ci_high": -0.0347222222222221, "p_value": 0.125, "rank_biserial": -0.866666666666667}, {"dataset_id": "Lee2019_MI", "n": 30, "median_delta_BACC": 0.01, "ci_low": -0.005, "ci_high": 0.025, "p_value": 0.2309220325143355, "rank_biserial": 0.2586206896551724}, {"dataset_id": "PhysioNetMI", "n": 55, "median_delta_BACC": 0.0, "ci_low": -0.0343477914915631, "ci_high": 0.0029761904761905, "p_value": 0.455185419014922, "rank_biserial": -0.1190130624092888}]}

Mandatory qualification: Not an A-number and not a replacement for primary A0.

Prohibited stronger wording: Do not claim the Stage11 challenger improved EEGNet generally.

Post-hoc/diagnostic status: DIAGNOSTIC_ONLY

P02-CLM-007/v1 - APPROVED_WITH_QUALIFICATIONS

Source candidate: P02-CLAIM-CAND-007

Reviewed wording: Within the canonical resource-constrained A4 scope, none of the 756 governed C1-C3 role-control statistical rows reached raw $p \leq 0.05$. This is an absence of statistically detected improvement under the executed scope, not proof that the true effect is zero and not evidence of five-repeat or dense-budget stability.

Evidence class: GOVERNED_ABLATION_NEGATIVE_NULL_RESULT

Claim ceiling: SUPPORTED_WITH_QUALIFICATION

Supporting findings: P02-FIND-009, P02-FIND-010

Numeric support: {"governed_C1_C3_statistical_rows": 756, "raw_p_le_0_05": 0, "scope": "single canonical refit repeat / governed deep anchor budgets"}

Mandatory qualification: Absence of detected effects is not proof of zero effect; canonical deep scope is reduced.

Prohibited stronger wording: Do not claim zero A4 effect, five-repeat stability, dense deep-budget characterization, or full-grid equivalence.

Post-hoc/diagnostic status: NO

P02-CLM-008/v1 - APPROVED_WITH_QUALIFICATIONS

Source candidate: P02-CLAIM-CAND-008

Reviewed wording: A hard-vote ensemble was not uniformly beneficial: one governed PhysioNetMI full-training C4 comparison at repeat 4 showed a Holm-significant disadvantage relative to the validation-selected EEGNet constituent (median delta BACC -0.0435; Holm $p=0.0371$). This cell-specific negative result does not establish that ensembles are generally inferior.

Evidence class: CELL_SPECIFIC_NEGATIVE_ABLATION_RESULT

Claim ceiling: SUPPORTED_WITH_QUALIFICATION

Supporting findings: P02-FIND-011

Numeric support: {"Holm_significant_rows": 1, "significant_rows": [{"comparison_id": "P02-A4-PhysioNetMI-ENSEMBLE-A4-C4-MODEL-HARD-VOTE-FULL_TRAIN-ER04:VS_ST", "dataset_id": "PhysioNetMI", "condition_id": "A4-C4-MODEL-HARD-VOTE", "budget_id": "FULL_TRAIN", "repeat": 4, "strongest_constituent_branch": "DNN-EEGNET", "closure_status": "COMPLETE", "n": 11.0, "median_delta_BACC": -0.0434782608695651, "p_value": 0.0185546875, "holm_adjusted_p": 0.037109375, "rank_biserial": -0.7878787878787878, "ci_low": -0.1304347826086956, "ci_high": -0.0119047619047618}]}}

Mandatory qualification: Do not generalize from one corrected comparison to all ensembles.

Prohibited stronger wording: Do not claim ensembles are generally inferior; one corrected comparison was negative.

Post-hoc/diagnostic status: NO

P02-CLM-009/v1 - APPROVED_WITH_QUALIFICATIONS

Source candidate: P02-CLAIM-CAND-009

Reviewed wording: Post-hoc Stage18S sensitivity evidence shows that A4 effect direction can change across descriptive repeats and budgets (22/45 anchor trajectories mixed-sign; all 15 six-budget MR00 trajectories changed sign at least once). This supports the existing claim ceiling but remains POST_HOC/DESCRIPTIVE/SENSITIVITY evidence and does not modify canonical Stage18/G18.

Evidence class: POST_HOC_SENSITIVITY

Claim ceiling: POST_HOC_ONLY

Supporting findings: P02-FIND-012

Numeric support: {"three_repeat_anchor_trajectories": 45, "all_negative": 7, "mixed": 22, "all_positive": 16, "six_budget_trajectories": 15, "six_budget_with_sign_flip": 15}

Mandatory qualification: Descriptive/post-hoc only; does not modify canonical G18.

Prohibited stronger wording: Do not present Stage18S as confirmatory or as modifying G18.

Post-hoc/diagnostic status: POST_HOC_DESCRIPTIVE_SENSITIVITY

P02-CLM-010/v1 - APPROVED_WITH_QUALIFICATIONS

Source candidate: P02-CLAIM-CAND-010

Reviewed wording: P02 provides a fail-closed, provenance-rich Layer-2 evidence substrate for downstream use while preserving successful and non-success outcomes, including 648 FailureCaseIndex records, 340 NegativeResultNotes and 309 DiagnosticOnlyFlags. This is technical evidence readiness, not deployment readiness or future-phase scientific success.

Evidence class: REPRODUCIBILITY_NEGATIVE_EVIDENCE

Claim ceiling: SUPPORTED

Supporting findings: P02-FIND-013, P02-FIND-014

Numeric support: {"FailureCaseIndex": 648, "NegativeResultNote": 340, "DiagnosticOnlyFlag": 309, "failure_terminal_breakdown": {"CONDITIONAL_SKIP": 591, "NOT_APPLICABLE_REPEAT_SLOT": 42, "INPUT_INCOMPATIBLE": 9, "FAILED": 3, "DEPENDENCY_BLOCKED": 3}}

Mandatory qualification: Technical readiness is not deployment readiness.

Prohibited stronger wording: Do not equate technical evidence readiness with deployment readiness.

Post-hoc/diagnostic status: NO

Cumulative P00-P02 reviewed claims

CUM-CLM-001/v1 - APPROVED_WITH_QUALIFICATIONS

Source candidate: CC-01

Reviewed wording: Through P02, IHARQ has a governed evidence chain linking validated foundation contracts, the frozen P01 provenance-traceable EEG substrate, and P02 Layer-2 decoder evidence through explicit identities, manifests and handoffs; this traceability does not by itself establish clinical or deployment effectiveness.

Supporting cumulative findings: CF-01, CF-02, CF-09

Supporting phases: P00, P01, P02

Claim ceiling: SUPPORTED_WITH_QUALIFICATION

Mandatory qualification: Technical traceability is not scientific/clinical effectiveness by itself.

Prohibited stronger wording: Do not state that the complete project is deployment-ready or clinically validated.

CUM-CLM-002/v1 - APPROVED_WITH_QUALIFICATIONS

Source candidate: CC-02

Reviewed wording: The P02 decoder analyses consume the frozen P01 subject-grouped, denominator-conserving binary motor-imagery data contract rather than a P02-local redefinition of labels, split or core windows, subject to the scope of the implemented registered integrity and leakage checks.

Supporting cumulative findings: CF-02

Supporting phases: P01, P02

Claim ceiling: SUPPORTED

Mandatory qualification: Bounded to the implemented registered leakage/disjointness checks.

Prohibited stronger wording: Do not claim that every conceivable leakage mechanism is impossible.

CUM-CLM-003/v1 - APPROVED_WITH_QUALIFICATIONS

Source candidate: CC-03

Reviewed wording: Across the three governed P02 datasets, decoder-family ordering is context-dependent rather than universally ordered, and participant heterogeneity on the multi-participant datasets limits any global winner claim; BNCI2014_001 is descriptive at the participant-inference level because its held-out P02 test set contains one participant.

Supporting cumulative findings: CF-03, CF-11

Supporting phases: P01, P02

Claim ceiling: SUPPORTED_WITH_QUALIFICATION

Mandatory qualification: BNCI held-out test participant n=1; no universal rank claim.

Prohibited stronger wording: Do not claim DBConformer, CSP-LDA or any other decoder is universally best.

CUM-CLM-004/v1 - APPROVED_WITH_QUALIFICATIONS

Source candidate: CC-04

Reviewed wording: The deterministic low-label populations prepared in P01 expose difficult, non-monotonic and rank-changing P02 behavior across the governed 1-32 labels/class surface; one frozen subset per budget prevents a stable repeated-subset sample-efficiency claim.

Supporting cumulative findings: CF-04

Supporting phases: P01, P02

Claim ceiling: SUPPORTED_WITH_QUALIFICATION

Mandatory qualification: Single frozen subset per budget; no repeated-subset stability.

Prohibited stronger wording: Do not claim stable low-label resilience or universal sample-efficiency superiority.

CUM-CLM-005/v1 - APPROVED_WITH_QUALIFICATIONS

Source candidate: CC-05

Reviewed wording: P01 resolved A4 data-feasibility by freezing a fully matched R2 substrate. Under P02's accepted constrained A4 execution, no generic C1-C3 statistically detected improvement was observed, and post-hoc Stage18S further shows repeat/budget sign sensitivity. These results do not establish zero effect, five-repeat stability, dense-budget behavior, or full original-grid equivalence.

Supporting cumulative findings: CF-05, CF-06

Supporting phases: P01, P02

Claim ceiling: SUPPORTED_WITH_QUALIFICATION

Mandatory qualification: Single canonical refit repeat, deep anchor budgets, Stage18S post-hoc.

Prohibited stronger wording: Do not claim A4 has zero effect, is five-repeat stable, or is fully equivalent to the originally planned dense grid.

CUM-CLM-006/v1 - APPROVED_WITH_QUALIFICATIONS

Source candidate: CC-06

Reviewed wording: The P02 correction history shows that apparent model viability under the implemented configurations depended on training-policy choice while the inherited P00/P01 data and identity foundations remained unchanged; this supports a reproduction/optimization-sensitivity interpretation, not universal architecture superiority or guaranteed gains from source-aligned settings.

Supporting cumulative findings: CF-07

Supporting phases: P00, P01, P02

Claim ceiling: SUPPORTED_WITH_QUALIFICATION

Mandatory qualification: Diagnostic/validation evidence does not establish universal test-set gains or architecture superiority.

Prohibited stronger wording: Do not claim the original uniform policy was theoretically invalid or that source-aligned settings universally improve every model.

CUM-CLM-007/v1 - APPROVED_WITH_QUALIFICATIONS

Source candidate: CC-07

Reviewed wording: Across P00-P02, fail-closed validation and explicit retention of failures, supersessions, null/negative results and diagnostic states provide an auditable project history in which non-success evidence remains available for interpretation; this process does not guarantee scientific correctness or deployment safety.

Supporting cumulative findings: CF-08, CF-09

Supporting phases: P00, P01, P02

Claim ceiling: SUPPORTED_WITH_QUALIFICATION

Mandatory qualification: Different negative-evidence classes are not quantitatively interchangeable.

Prohibited stronger wording: Do not claim that fail-closed process guarantees scientific correctness or deployment safety.

CUM-CLM-008/v1 - APPROVED_WITH_QUALIFICATIONS

Source candidate: CC-08

Reviewed wording: At the end of P02, canonical Layer-2 evidence is technically ready for governed downstream consumption, including P03, while explicit scope, repeat/stability, access and non-clinical limitations remain mandatory handoff constraints; this does not imply future-phase scientific success.

Supporting cumulative findings: CF-12

Supporting phases: P01, P02

Claim ceiling: SUPPORTED

Mandatory qualification: Downstream science is not yet executed.

Prohibited stronger wording: Do not equate technical handoff readiness with future scientific success.

Historical P00/P01 preservation and P02 effect

- P00-CLM-001/v2 - historical disposition APPROVED_WITH_QUALIFICATIONS preserved; P02 effect NO_CHANGE; current state ACTIVE_QUALIFIED_CURRENT.
- P00-CLM-002/v2 - historical disposition APPROVED_WITH_QUALIFICATIONS preserved; P02 effect NO_CHANGE; current state ACTIVE_QUALIFIED_CURRENT.
- P00-CLM-003/v2 - historical disposition APPROVED_WITH_QUALIFICATIONS preserved; P02 effect NO_CHANGE; current state ACTIVE_QUALIFIED_CURRENT.
- P00-CLM-004/v2 - historical disposition APPROVED_WITH_QUALIFICATIONS preserved; P02 effect NO_CHANGE; current state ACTIVE_QUALIFIED_CURRENT.
- P00-CLM-005/v2 - historical disposition APPROVED_WITH_QUALIFICATIONS preserved; P02 effect NO_CHANGE; current state ACTIVE_QUALIFIED_CURRENT.
- P00-CLM-006/v2 - historical disposition APPROVED_WITH_QUALIFICATIONS preserved; P02 effect NO_CHANGE; current state ACTIVE_QUALIFIED_CURRENT.
- P00-CLM-007/v2 - historical disposition APPROVED_WITH_QUALIFICATIONS preserved; P02 effect NO_CHANGE; current state ACTIVE_QUALIFIED_CURRENT.
- P01-CLM-001/v1 - historical disposition APPROVED_WITH_QUALIFICATIONS preserved; P02 effect NO_CHANGE; current state ACTIVE_HISTORICAL.
- P01-CLM-002/v1 - historical disposition APPROVED_WITH_QUALIFICATIONS preserved; P02 effect NO_CHANGE; current state ACTIVE_HISTORICAL.
- P01-CLM-003/v1 - historical disposition APPROVED preserved; P02 effect NO_CHANGE; current state ACTIVE_HISTORICAL.
- P01-CLM-004/v1 - historical disposition APPROVED_WITH_QUALIFICATIONS preserved; P02 effect NO_CHANGE; current state ACTIVE_HISTORICAL.
- P01-CLM-005/v1 - historical disposition APPROVED_WITH_QUALIFICATIONS preserved; P02 effect NO_CHANGE; current state ACTIVE_HISTORICAL.
- P01-CLM-006/v1 - historical disposition APPROVED_WITH_QUALIFICATIONS preserved; P02 effect STRENGTHENING_EVIDENCE; current state ACTIVE_QUALIFIED_WITH_P02_REALIZED_DOWNSTREAM_USE.
- P01-CLM-007/v1 - historical disposition APPROVED_WITH_QUALIFICATIONS preserved; P02 effect STRENGTHENING_EVIDENCE; current state ACTIVE_QUALIFIED_WITH_P02_REALIZED_DOWNSTREAM_USE.
- P01-CLM-008/v1 - historical disposition APPROVED preserved; P02 effect NO_CHANGE; current state ACTIVE_HISTORICAL.
- P01-CLM-009/v1 - historical disposition APPROVED_WITH_QUALIFICATIONS preserved; P02 effect NO_CHANGE; current state ACTIVE_HISTORICAL.
- P01-CLM-010/v1 - historical disposition APPROVED_WITH_QUALIFICATIONS preserved; P02 effect STRENGTHENING_EVIDENCE; current state ACTIVE_QUALIFIED_WITH_P02_REALIZED_DOWNSTREAM_USE.
- P01-CLM-011/v1 - historical disposition DEFERRED preserved; P02 effect SUPERSESSION; current state HISTORICAL_DEFERRED_CLAIM_RESOLVED_BY_P02_GENERIC_IMPROVEMENT_N
- P01-CLM-012/v1 - historical disposition REJECTED preserved; P02 effect STRENGTHENING_EVIDENCE; current state REJECTED_SCOPE_BOUNDARY_REMAINS_ACTIVE_THROUGH_P02.

Layer 0 closure gate

- Every material P02 candidate claim reviewed: **PASS**.
- Every cumulative candidate claim reviewed: **PASS**.
- Major limitations mapped: **PASS**.
- Stage18S post-hoc boundary encoded: **PASS**.
- A4 resource claim ceiling encoded: **PASS**.
- Prohibited stronger wording registered: **PASS**.
- Measurements/Protocol/execution changed by Layer 0: **NO**.

LAYER0_THROUGH_P02_STATUS = FINALIZED